Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

CompTIA Security+ Certification (SY0-701)

TGS Ref No: TGS-2023039181

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	2 July 2026	Initial release — Security+ SY0-701 3-day lesson plan aligned to the 36 labs.	Dr. Alfred Ang

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Course Information

Course Title	CompTIA Security+ Certification (SY0-701)
WSQ Course Reference	TGS-2023039181
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	3 days · 8 training hours per day (24 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led, hands-on labs on the free Killercoda Ubuntu Playground
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Apply general security concepts — control categories and types, the CIA triad and AAA, cryptographic primitives (hashing, symmetric/asymmetric encryption) and PKI.
- LO2: Identify threats, vulnerabilities and mitigations — phishing, password and web attacks (SQLi, XSS), malware, denial-of-service, vulnerability scanning and host hardening.
- LO3: Implement secure architecture — network segmentation, firewalls, WAF, VPN, data-at-rest encryption, TLS hardening and resilient backup.
- LO4: Run security operations — SSH/identity hardening, MFA, SIEM, IDS, EDR, file-integrity monitoring, vulnerability management, incident response and automation.
- LO5: Manage the security program — risk registers and CVSS, reconnaissance, penetration testing, compliance auditing and security-awareness testing.

Assessment

- Written Assessment (WA) — Short-Answer Questions (SAQ), 6 questions, 1 hour, open book.
- Practical Performance (PP) — 5 hands-on security tasks, 1 hour, open book.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- Final assessment is conducted on Day 3 from 4:30 pm.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — General Security Concepts & Threats

Time	Duration	Topic / Activity
9:30–10:00	30 min	Welcome, course introduction, ground rules and mandatory digital attendance (AM)
10:00–11:00	60 min	**Domain 1 — General Security Concepts: control categories**

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		& types, CIA triad & AAA, hashing, symmetric/asymmetric encryption and PKI (concepts + demo)
11:00–11:15	15 min	Tea break
11:15–13:00	105 min	Hands-on: Lab 1: Security Controls Categorization; Lab 2: CIA Triad and AAA on Linux; Lab 3: Hashing, Salting, and Digital Signatures; Lab 4: Symmetric vs. Asymmetric Encryption; Lab 5: PKI: Build Your Own Certificate Authority
13:00–14:00	60 min	Lunch break
14:00–14:45	45 min	**Domain 2 — Threats, Vulnerabilities & Mitigations: social engineering, malware, web & password attacks, scanning and hardening (concepts)**
14:45–16:15	90 min	Hands-on: Lab 6: Phishing Email Header Analysis (SPF / DKIM / DMARC); Lab 7: Password Attacks: Spraying, Brute Force, Cracking; Lab 8: Malware Static Analysis (Safe Sample); Lab 9: SQL Injection (SQLi)
16:15–16:30	15 min	Tea break
16:30–18:15	105 min	Hands-on: Lab 10: Cross-Site Scripting (XSS); Lab 11: DoS Simulation and Detection; Lab 12: Vulnerability Scanning with Nikto and Nuclei; Lab 13: Linux Hardening Baseline with Lynis
18:15–18:30	15 min	Day 1 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 2 — Security Architecture & Security Operations

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 recap and mandatory digital attendance (AM)
9:45–10:30	45 min	**Domain 3 — Security Architecture: segmentation, firewalls, WAF, VPN, encryption at rest, TLS and backup (concepts + demo)**
10:30–10:45	15 min	Tea break
10:45–13:00	135 min	Hands-on: Lab 14: Network Segmentation with Linux Namespaces; Lab 15: Stateful Firewall with UFW and nftables; Lab 16: Web Application Firewall with ModSecurity; Lab 17: Site-to-

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		Site VPN with WireGuard; Lab 18: Data at Rest: LUKS Full-Disk Encryption
13:00–14:00	60 min	Lunch break
14:00–14:30	30 min	**Domain 4 — Security Operations: hardening, IAM/MFA, SIEM, IDS, EDR, FIM, vuln management, IR and automation (concepts)**
14:30–16:15	105 min	Hands-on: Lab 19: TLS Hardening and Cipher Audit; Lab 20: Backup and Recovery with restic; Lab 21: SSH Hardening + Fail2ban; Lab 22: IAM with PAM and sudo; Lab 23: Multi-Factor Authentication with TOTP
16:15–16:30	15 min	Tea break
16:30–18:15	105 min	Hands-on: Lab 24: SIEM Basics with Wazuh; Lab 25: Network IDS with Suricata; Lab 26: Endpoint Detection with osquery
18:15–18:30	15 min	Day 2 recap, Q&A and PM digital attendance

Total training time: 480 minutes (8 hours).

Day 3 — Security Operations, Program Management & Assessment

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 recap and mandatory digital attendance (AM)
9:45–11:15	90 min	Hands-on: Lab 27: File Integrity Monitoring with AIDE; Lab 28: Vulnerability Management with Trivy; Lab 29: Incident Response and Forensics; Lab 30: Automation and Orchestration (Mini-SOAR)
11:15–11:30	15 min	Tea break
11:30–13:00	90 min	**Domain 5 — Security Program Management & Oversight: risk & CVSS, reconnaissance, pentesting, compliance and awareness (concepts). Hands-on: Lab 31: Risk Register, CVSS Scoring, and Quantitative Risk; Lab 32: Passive Reconnaissance / OSINT**
13:00–14:00	60 min	Lunch break
14:00–16:00	120 min	Hands-on: Lab 33: Active Reconnaissance with Nmap; Lab 34: Penetration Testing with Metasploit (Authorised Lab); Lab 35: Compliance Audit with OpenSCAP; Lab 36: Security Awareness: Run a

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		Phishing Simulation with GoPhish
16:00–16:15	15 min	Tea break
16:15–16:30	15 min	**Briefing for Assessment**
16:30–17:30	60 min	**Written Assessment (WA) — Short-Answer Questions (SAQ), 6 questions, 1 hour, open book**
17:30–18:30	60 min	**Practical Performance (PP) — 5 hands-on security tasks, 1 hour, open book. PM digital attendance**

Total training time: 480 minutes (8 hours).

Lab Reference (aligned to exam domains)

Domain / exam area	Weighting	Labs
Domain 01: General Security Concepts	12%	Lab 1, Lab 2, Lab 3, Lab 4, Lab 5
Domain 02: Threats, Vulnerabilities & Mitigations	22%	Lab 6, Lab 7, Lab 8, Lab 9, Lab 10, Lab 11, Lab 12, Lab 13
Domain 03: Security Architecture	18%	Lab 14, Lab 15, Lab 16, Lab 17, Lab 18, Lab 19, Lab 20
Domain 04: Security Operations	28%	Lab 21, Lab 22, Lab 23, Lab 24, Lab 25, Lab 26, Lab 27, Lab 28, Lab 29, Lab 30
Domain 05: Security Program Management & Oversight	20%	Lab 31, Lab 32, Lab 33, Lab 34, Lab 35, Lab 36

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